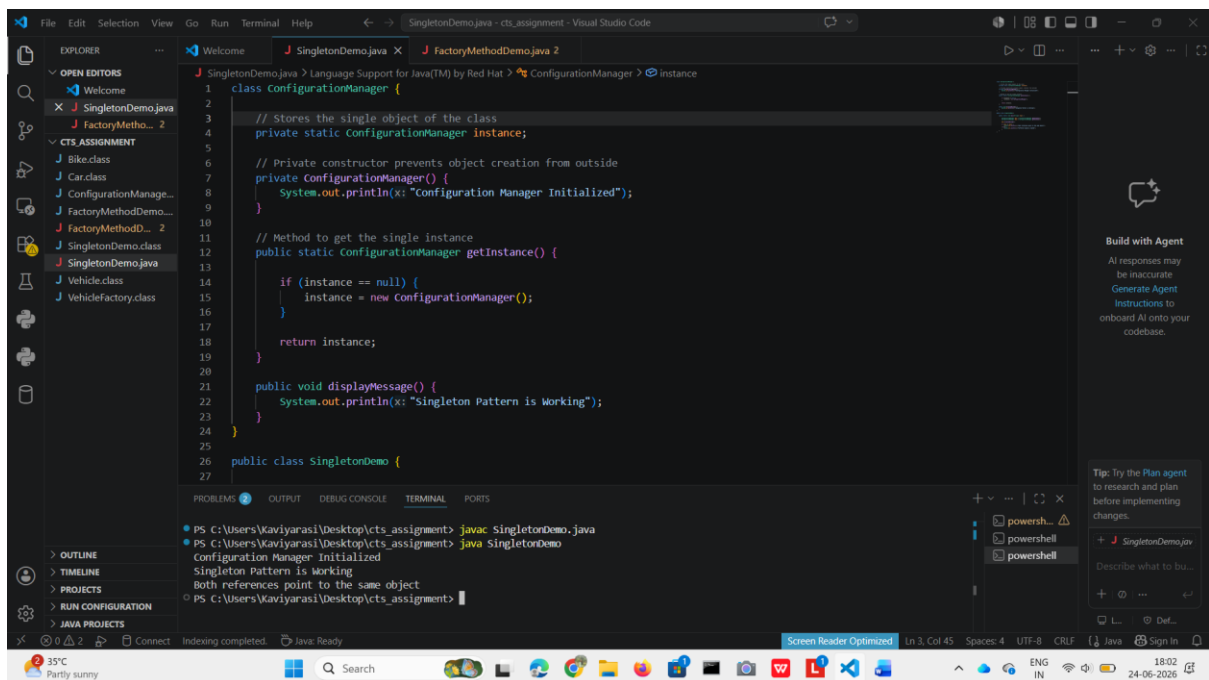


CTS DEEPSKILLING ASSIGNMENTS (OUTPUT) – WEEK 1

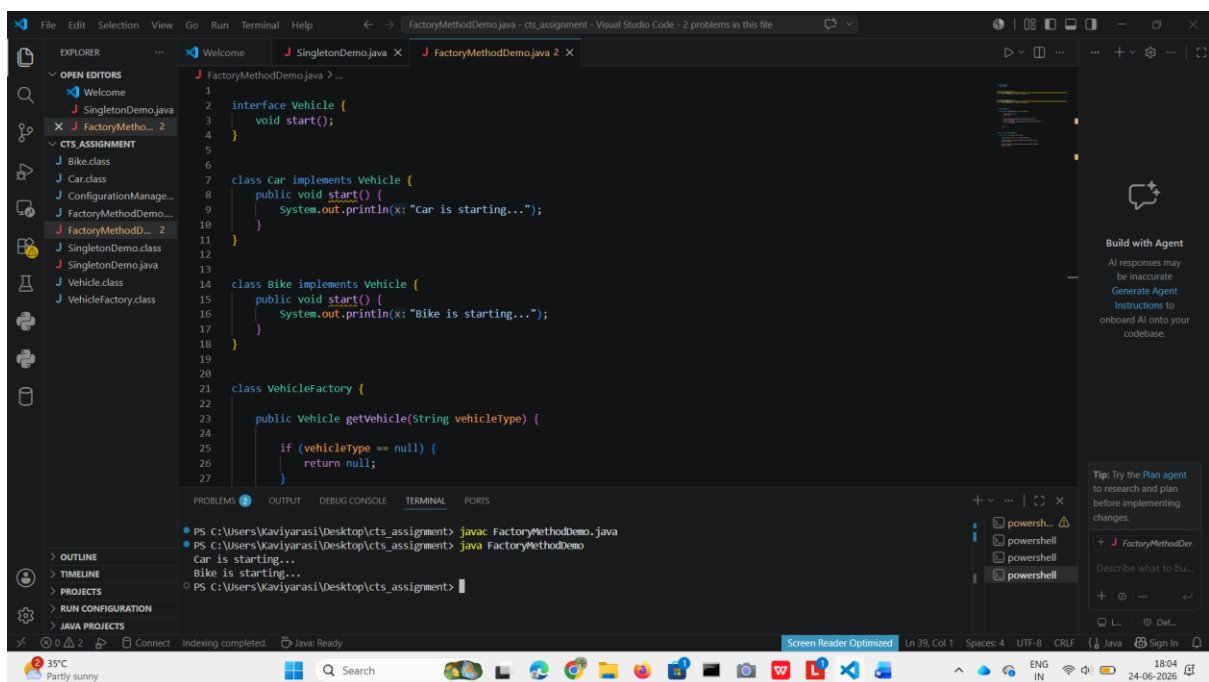
1) SINGLETON PATTERN -DESIGN PATTERN AND PRINCIPLE



```
1 class ConfigurationManager {
2     // Stores the single object of the class
3     private static ConfigurationManager instance;
4
5     // Private constructor prevents object creation from outside
6     private ConfigurationManager() {
7         System.out.println(x: "Configuration Manager Initialized");
8     }
9
10    // Method to get the single instance
11    public static ConfigurationManager getInstance() {
12        if (instance == null) {
13            instance = new ConfigurationManager();
14        }
15        return instance;
16    }
17
18    public void displayMessage() {
19        System.out.println(x: "Singleton Pattern is Working");
20    }
21
22    public class SingletonDemo {
23
24    }
25
26    }
27
```

PS C:\Users\Kaviyarasi\Desktop\cts_assignment> javac SingletonDemo.java
PS C:\Users\Kaviyarasi\Desktop\cts_assignment> java SingletonDemo
Configuration Manager Initialized
Singleton Pattern is Working
Both references point to the same object
PS C:\Users\Kaviyarasi\Desktop\cts_assignment>

2) FACTORY METHOD PATTERN -DESIGN PATTERN AND PRINCIPLE



```
1 interface Vehicle {
2     void start();
3 }
4
5 class Car implements Vehicle {
6     public void start() {
7         System.out.println(x: "Car is starting...");
8     }
9 }
10
11 class Bike implements Vehicle {
12     public void start() {
13         System.out.println(x: "Bike is starting...");
14     }
15 }
16
17 class VehicleFactory {
18
19     public Vehicle getVehicle(String vehicleType) {
20         if (vehicleType == null) {
21             return null;
22         }
23     }
24
25 }
26
27
```

PS C:\Users\Kaviyarasi\Desktop\cts_assignment> javac FactoryMethodDemo.java
PS C:\Users\Kaviyarasi\Desktop\cts_assignment> java FactoryMethodDemo
Car is starting...
Bike is starting...
PS C:\Users\Kaviyarasi\Desktop\cts_assignment>